

Scalability & Future Plan

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This document outlines how the Weather & Location Analytics Backend can be extended to support larger user bases, more weather data, additional features and future AI capabilities.

Current System Limitations:

Scalability Plan:

Future Roadmap:

Date	15 March 2024		
Team ID	6ab4d2876603573ccfb56348		
Project Name	Weather & Location Analytics Backend		
Maximum Marks	1 Mark		

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	External API Dependency	Weather information depends on availability of the third-party weather service.	High
2	Current Feature Scope	The current project focuses on core weather, location, authentication and AI recommendation features.	Medium
3	Growth of User/Data Volume	More users and saved locations may require stronger database and performance management.	High

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Backend supports current project users.	Use scalable server deployment and efficient request handling.
2	Data Storage	MongoDB stores user information.	Expand database capacity and indexing as data grows.
3	Performance	Express.js handles API requests and external service communication.	Add caching, optimized routes and monitoring when required.
4	Security	Authentication and environment-based configuration are included.	Strengthen JWT/session security, access controls and secret management.

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Improve authentication, account management and API security.	Near-term	Stronger user access control.
Phase 2	Enhance AI contextual recommendations and weather summaries.	Future	More useful daily planning support.
Phase 3	Add push notifications and offline access.	Future	Improved accessibility and alerts.
Phase 4	Extend to mobile platforms and advanced weather analytics.	Long-term	Broader access and weather trend analysis.